

Stochastic Programming.
Problems session 4

15. A bank branch wants to determine the amount of money to be deposited in an ATM on Fridays, before the weekend. The branch estimates that money has a cost (associated to the loss of benefit by interest rates) of $R \text{ €}$ for each € in the ATM. The demand of money during the weekend is a random variable ξ with continuous uniform distribution at interval $m \leq \xi \leq M$.

The ATM has a capacity of $L \text{ €}$, where $L \in [m, M]$. We will also assume that x has a lower bound m . If the demand exceeds the amount x deposited in the ATM, it generates a fixed cost of $C \text{ €}$. Consider that a two-stage stochastic problem is formulated, using a second stage function that verifies

$$Q(x, \xi) = \begin{cases} 0 & \text{if } x \geq \xi \\ C & \text{if } x < \xi. \end{cases}$$

- (a) Compute analytically the optimal solution of previous problem, computing first the recourse function $Q(x) = E_{\xi}[Q(x, \xi)]$.
- (b) Compute the Expected value of Perfect information EVPI, and the value of stochastic solution VSS for this problem. Take the particular values: $R = 0.09$, $L = 100$, $m = 10$, $M = 110$, $C = 10$.
16. To plan the Catalan Easter Monday (“dia de la mona”), a bakery presents a stochastic problem where x is the number of “mones” (Catalan Easter cake) that will cook in the first stage. The cooking cost for a “mona” is 10 € , and the maximum number of “mones” that can cook and/or store is 1000. The demand of “mones” in this bakery is a random variable ξ that follows a continuous uniform distribution at interval $[500, 2000]$. The bakery formulates the following stochastic problem, where v is the selling price of every “mona”.

$$\begin{array}{ll} \min_x & 10x + E_{\xi}[Q(x, \xi)] \\ \text{s.to} & 500 \leq x \leq 1000 \end{array} \quad \text{where} \quad \begin{array}{ll} Q(x, \xi) = \min_y & -vy \\ \text{s.to} & y \leq \xi \\ & y \leq x \\ & y \geq 0 \end{array}$$

- (a) Obtain the sets K_1 and K_2 of this problem.
- (b) Compute the selling price v such that the solution of the problem is 750 “mones”.
- (c) Assuming that the bakery fixes the selling price v to 12.5 € , compute the Expected value of perfect information EVPI.
17. A company wants to invest an amount of money in n financial assets (share, bonds, etc.). It wants to decide which fraction to invest in each asset $i = 1, \dots, n$. Consider a simplification of this problem with only two assets. Let x_1 and x_2 be the fractions invested in asset 1 and 2. Acquiring assets has expenses of 2% and 1% for assets 1 and 2, respectively.

The future returns of second asset is 0.12%, while for the first asset is a random variable that follows a continuous uniform distribution at interval $[0.10, 0.20]$. The company wants to reach a global future returns of 0.13% at the end of period. If it falls below, it means a cost of 2.5% of the total invested money. Solve the next problem formulated by the company:

$$\begin{array}{ll} \min_{x_1, x_2} & 0.02x_1 + 0.01x_2 + E_{\xi_1}[Q(x, \xi_1)] \\ \text{s.to} & x_1 + x_2 = 1 \\ & 0 \leq x_1, x_2 \leq 1 \end{array} \quad \text{where} \quad Q(x, \xi_1) = \begin{cases} 0 & \text{if } \xi_1 x_1 + 0.12x_2 \geq 0.13 \\ 0.025 & \text{if } \xi_1 x_1 + 0.12x_2 < 0.13. \end{cases}$$

Hint: work only with x_1 , removing x_2 through $x_1 + x_2 = 1$.

- (a) Compute the Expected value of perfect information EVPI.
 - (b) Compute the Value of stochastic solution VSS.
18. A company wants to collect a sample of size n for some study. Collecting a datum of the sample has a cost of c €, and it requires two hours of work to obtain it. We have a total of 300 available hours. If once collected the sample and started the study, we realize that more data would be required, we have a cost of 100 € for every missing data.

The number of needed data is a random variable ξ . To simplify, we also assume that n is continuous. The company formulates the following stochastic problem:

$$\begin{array}{llll} \min & cn + E_{\xi}[Q(n, \xi)] & \text{where} & Q(x, \xi) = \min_{y \geq 0} 100y \\ \text{s.to} & 2n \leq 300 & \text{where} & \text{s.a} \quad n + y \geq \xi \\ & n \geq \xi_m & & y \geq 0, \end{array}$$

being ξ_m the minimum value of the random variable ξ .

Assignment

- (a) Considering that ξ has a continuous uniform distribution at interval $[100, 200]$, compute the value of c such that the solution is $n = 120$.
- (b) Considering again that ξ has a continuous uniform distribution at interval $[100, 200]$, compute the Expected value of perfect information EVPI and the Value of stochastic solution VSS, taking $c = 80$.